Problem 8.5: Flow rate is $Q = 15\ m^3/s$

a) Draw specific diagram using given data.

b) Explain the diagram and determine critical depth.

c) Define specific energy, total head and write the relation between them. What are the unit definitions for both concepts? Define the critical regime for two different ways. Show a discharge will be found for two different depths with existing specific energy using for Q=constant, E=E(y) or for E= constant, Q=Q(y) , and critical regime will be found for their limit cases.

d) When specific energy is constant for uniform flow in a rectangular channel, is it possible energy loss through the channel? Explain.

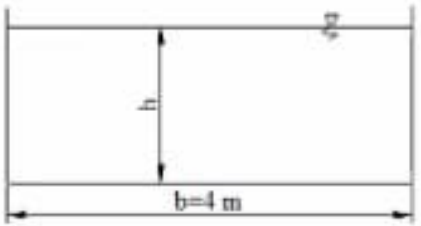


Figure p8.5(a)

Solution:

$$E = \frac{V^2}{2g} + h = \frac{Q^2}{2gA^2} + h$$

$$A = bh = 4h$$

$$E = \frac{15^2}{2g(4h)^2} + h = \frac{0.717}{h^2} + h$$

H	0.5	0.75	1	1.1	1.128	1.25	1.5
E	3.368	2.025	1.717	1.693	1.692	1.702	1.819

$$h_c = \sqrt[3]{\frac{Q^2}{B^2.g}} = \sqrt[3]{\frac{15^2}{16 \times 9.81}} = 1.128 \text{ m}$$

$$E_{min} = \frac{3}{2} h_c = 1.692$$

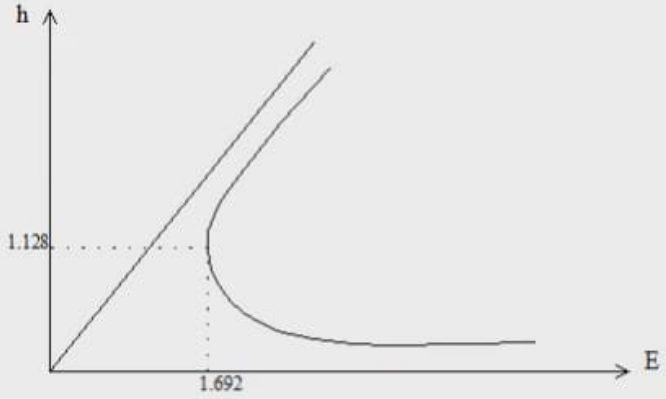


Figure p8.4(b) Specific energy diagram

c) Specific energy = $E = y + \frac{V^2}{2g}$

Total energy = $H = z + y + \frac{V^2}{2g}$

$H = z + E$

Both terms were written for unit weight.

Critical flow

1) For constant discharge and minimum specific energy.

2) For constant specific energy and maximum discharge

For Q=constant, the function of $E=E(y)$

For $E = E_{min}$, single solution as $y = y_c$

Two solutions for $E > E_{min}$

$y = y_1 < y_c$ supercritical

Two solutions for $E > E_{min}$

$y = y_1 < y_c$ supercritical

$y = y_2 > y_c$ subcritical

When $E < E_{min}$, Q does not flow with existing E , E should increase

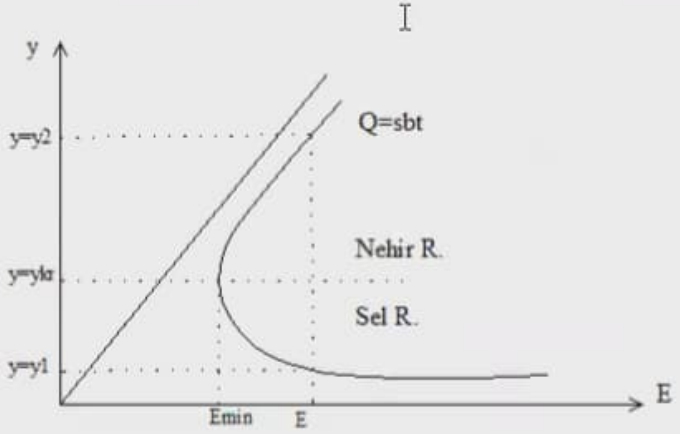


Figure p8.5(a)

d) When $E = y + \frac{V^2}{2g} = \text{constant}$, at distance Δx, there is energy loss as $I_0 \Delta x$. Since flow is uniform and steady; $I_e = I_0$ and so.

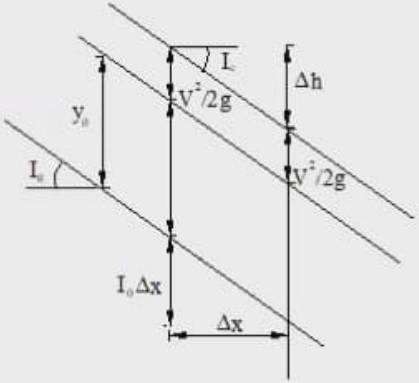


Figure p8.5(b)

Problem 8.6: Uniform flow depth is 3m, velocity is 2m/s and assume the channel is wide.

a) Determine the flow regime.

b) A sill constructed on the bed has 25cm height. Show, is it possible the flow past over the sill? What is the water depth on the sill?

c) What is maximum (critical) height of the sill?

Solution:

a) For rectangular channel

$$y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{36}{9.81}} = 1.54 < 3 \text{ m}$$ flow is in sub-critical regime.

$q = 3 \times 2 = 6 \text{ m}^3/\text{sm}$

or

$$Fr = \frac{V}{\sqrt{gy}} = \frac{2}{\sqrt{9.81 \times 3}} = \frac{2}{5.42} = 0.37 < 1$$

Problem 8.10: 15 m^3/s flowrate flows in a channel shown in Fig. 8.10. Assuming the roughness coefficient for the side channel and main channel n= 0.035. Side slopes are 1:2 and width of the channel is 5 m. Calculate the discharge in the channel. Bed slope is 10 m/km at the upstream section and it changes to 50 m/km at the downstream section.

(a) Find uniform water depth at the upstream and downstream section.

(b) Find the critical water depth at the upstream and downstream section.

(c) Find the Froude number at the upstream and downstream section. Determine the regimes.

(d) Draw water surface level.

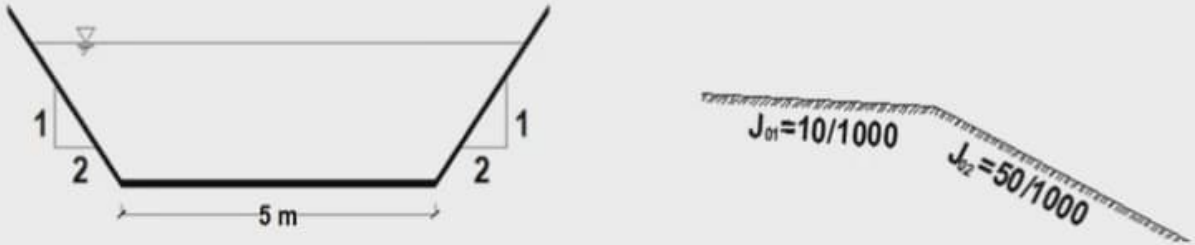


Figure 8.10

$$A = 5h + 2h^2 \qquad P = 2\sqrt{5}h + 5 \qquad R = \frac{5h + 2h^2}{2\sqrt{5}h + 5}$$

a) Menbadaki Üniform Akım Derinliği:

$$15 = \frac{1}{0.035} \times \left[\frac{5h + 2h^2}{2\sqrt{5}h + 5}\right]^{2/3} \times (0.01)^{1/2} \times \left(5h + 2h^2\right) \Rightarrow h = 0.953 \ m.$$

b) Mansaptaki Üniform Akım Derinliği:

$$15 = \frac{1}{0.035} \times \left[\frac{5h + 2h^2}{2\sqrt{5}h + 5}\right]^{2/3} \times (0.05)^{1/2} \times \left(5h + 2h^2\right) \Rightarrow h = 0.61 \ m.$$

c) Kritik Akış Derinliği:

$$\frac{Q^2}{g}=\frac{A^3}{B}\Rightarrow\frac{15^2}{9.81}=\frac{\left[5y_{kr}+2y_{kr}^2\right]^3}{\left[5+4y_{kr}\right]}\Rightarrow y_{kr}=0.86\ m.$$

$$V_1=\frac{15}{\left[5\times0.953+2\times0.953^2\right]}=2.28\ m/s\rightarrow y_{m1}=\frac{\left[5\times0.953+2\times0.953^2\right]}{\left[5+4\times0.953\right]}=0.747\ m.$$

$$Fr_1=\frac{V_1}{\sqrt{gy_{m1}}}=\frac{2.28}{\sqrt{9.81\times0.747}}\rightarrow Fr_1=0.84 \qquad Fr_1<1 \text{ Nehir Rejimi}$$

$$V_2=\frac{15}{\left[5\times0.61+2\times0.61^2\right]}=3.95\ m/s\rightarrow y_{m2}=\frac{\left[5\times0.61+2\times0.61^2\right]}{\left[5+4\times0.61\right]}=0.51\ m.$$

$$Fr_2=\frac{V_2}{\sqrt{gy_{m2}}}=\frac{3.95}{\sqrt{9.81\times0.51}}\rightarrow Fr_1=1.77 \qquad Fr_2>1 \quad \text{Sel Rejimi}$$